

# SOFTWARE ARCHITECTURE DOCUMENT

*PRISM*

*AI-Powered Automotive Procurement & Supplier Risk Management System*

Academic Year: 2026–2027

## 1. INTRODUCTION

### 1.1 PURPOSE

This Software Architecture Document defines the technical architecture of PRISM, an AI-powered automotive procurement and supplier risk intelligence system.

It describes:

- Major architectural components
- Component responsibilities
- Communication and data flow
- AI-agent orchestration
- Deployment structure
- Security and quality attributes

PRISM follows a modular architecture where frontend, backend, AI orchestration, ML services and database responsibilities are clearly separated.

### 1.2 SCOPE

The architecture supports:

- Procurement planning
- Supplier management
- RFQ management
- Quotation processing
- Supplier intelligence
- Quotation comparison
- Historical price intelligence
- Risk prediction and monitoring

- Decision support
- Negotiation assistance
- Human approval
- Purchase-order execution
- Continuous monitoring

## 1.3 KEY DEFINITIONS

| TERM  | MEANING                                             |
|-------|-----------------------------------------------------|
| PRISM | Procurement Risk Intelligence & Supplier Management |
| SAD   | Software Architecture Document                      |
| PRD   | Product Requirements Document                       |
| RFQ   | Request for Quotation                               |
| BOM   | Bill of Materials                                   |
| PO    | Purchase Order                                      |
| API   | Application Programming Interface                   |
| ML    | Machine Learning                                    |
| AI    | Artificial Intelligence                             |
| RBAC  | Role-Based Access Control                           |
| OCR   | Optical Character Recognition                       |
| LLM   | Large Language Model                                |

## 2. ARCHITECTURAL GOALS & CONSTRAINTS

### 2.1 ARCHITECTURAL GOALS

#### Modularity

Separate major responsibilities into independent components and workflows.

#### Separation of Responsibilities

Each technology performs its intended role:

- React/Vite → Presentation
- FastAPI → Backend/API
- SNS Workbench → Agent orchestration
- Python → ML and analytics
- PostgreSQL → Structured data

### **Proactive Intelligence**

Support continuous analysis of supplier and procurement signals.

### **Explainability**

Risk assessments and recommendations should provide evidence and reasoning.

### **Human Control**

Important procurement and commercial decisions remain under human control.

### **Scalability**

Allow additional suppliers, components, transactions, models and workflows without redesigning the complete system.

## **2.2 CONSTRAINTS**

- Automotive procurement focus
- SNS Agent Workbench for AI orchestration
- Python for ML/analytics
- PostgreSQL as primary database
- React + Vite frontend
- FastAPI backend
- RBAC
- No autonomous final commercial decisions
- No fabricated supplier history
- Prototype may use realistic synthetic procurement data

These goals and constraints are explicitly defined in the current SAD.

## **3. ARCHITECTURAL PRINCIPLES**

PRISM follows these architectural principles:

1. **Separation of Concerns**
2. **Modular Design**
3. **API-Based Communication**
4. **Data-Driven Intelligence**
5. **Human-in-the-Loop**
6. **Explainable AI**
7. **Secure Access**
8. **Event-Driven Monitoring**
9. **Scalable Services**
10. **Auditable Decisions**

## **ARCHITECTURE DECISION**

Instead of creating one large AI system, PRISM separates procurement intelligence into specialized workflows coordinated by a Master Procurement Agent.

### **PRINCIPLE**

**Specialized responsibility + centralized orchestration + human approval**

This allows individual workflows to evolve independently while maintaining a common procurement process.

## **4. SYSTEM CONTEXT**

PRISM operates between procurement stakeholders and procurement-related data sources.

### **PRIMARY USER**

**Procurement Manager**

### **Other Users**

- Procurement Buyer / Analyst
- Supplier
- Operations Team
- Finance / Approver
- Production / Supply Chain Team

### **External Systems**

- ERP systems
- Procurement data sources
- Market/reference data sources

## CONTEXT FLOW

### Users / External Systems

↓

### PRISM

↓

### Procurement Intelligence

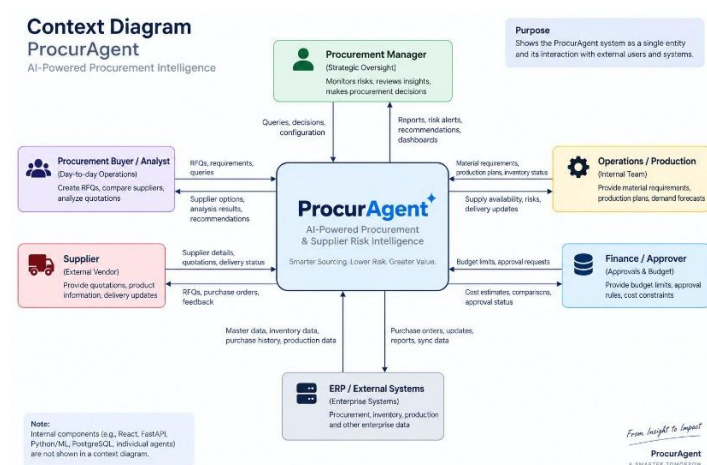
↓

### Insights / Recommendations

↓

### Human Decision

The Procurement Manager can review requirements, create RFQs, compare suppliers, review risks, view recommendations, negotiate and monitor procurement status.



## 5. HIGH-LEVEL ARCHITECTURE

### 5.1 ARCHITECTURAL LAYERS

## **1. Presentation Layer**

### **React + Vite**

Dashboards, supplier management, RFQs, quotation comparison, risk insights, negotiation and approvals.

## **2. Application/API Layer**

### **FastAPI**

Authentication, authorization, APIs, business services, routing, database access and integrations.

## **3. AI Orchestration Layer**

### **SNS Agent Workbench**

Agent coordination, context passing, event handling, workflow execution and human-in-the-loop tasks.

## **4. Intelligence Layer**

### **Python**

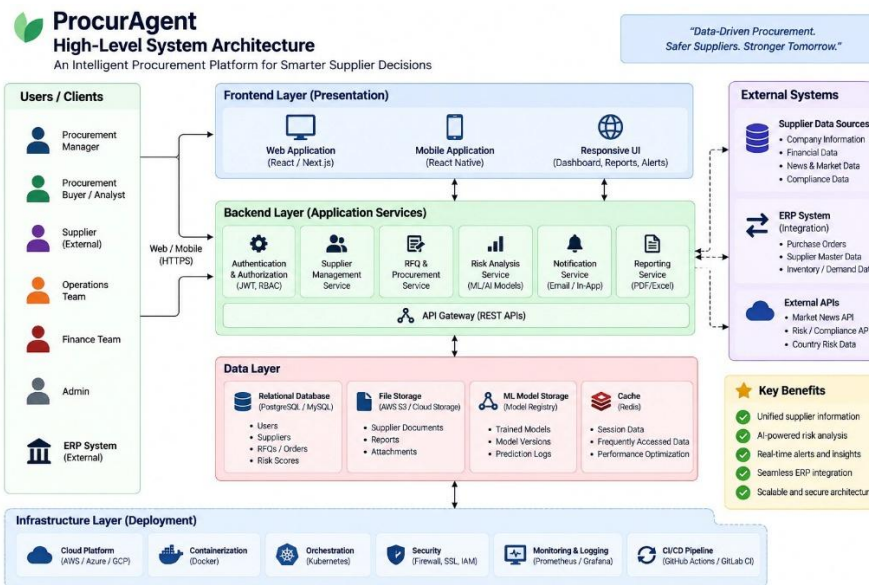
ML models, prediction, forecasting, risk scoring, price analysis and analytical processing.

## **5. Data Layer**

### **PostgreSQL**

Structured procurement and supplier data.

The current SAD defines these five layers and their responsibilities.



## 6. DATA FLOW & TECHNOLOGY INTERACTION

### 6.1 High-Level Data Flow

#### Users / External Systems



#### React + Vite



#### FastAPI API Gateway



#### SNS Agent Workbench



#### Python ML / Analytics



#### PostgreSQL

The processed information returns through the backend to the frontend as:

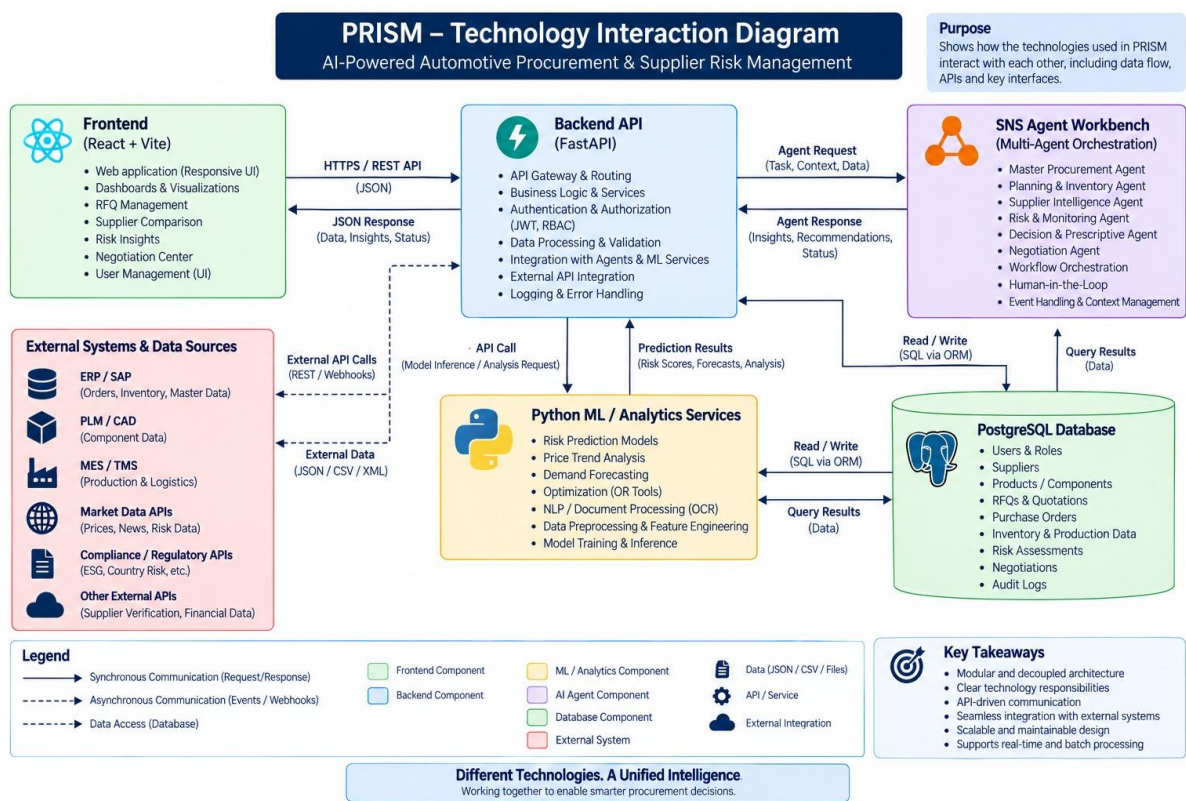
- Dashboards
- Risk alerts
- Supplier comparisons
- Recommendations

External procurement and market information may enter through the integration layer.

## 6.2 TECHNOLOGY INTERACTION

### Component Communicates With Purpose

|         |                   |                          |
|---------|-------------------|--------------------------|
| React   | FastAPI           | User/API requests        |
| FastAPI | PostgreSQL        | Data access              |
| FastAPI | SNS               | Workflow execution       |
| SNS     | Python            | ML/analytical processing |
| SNS     | External services | Required integrations    |
| FastAPI | External systems  | Data integration         |



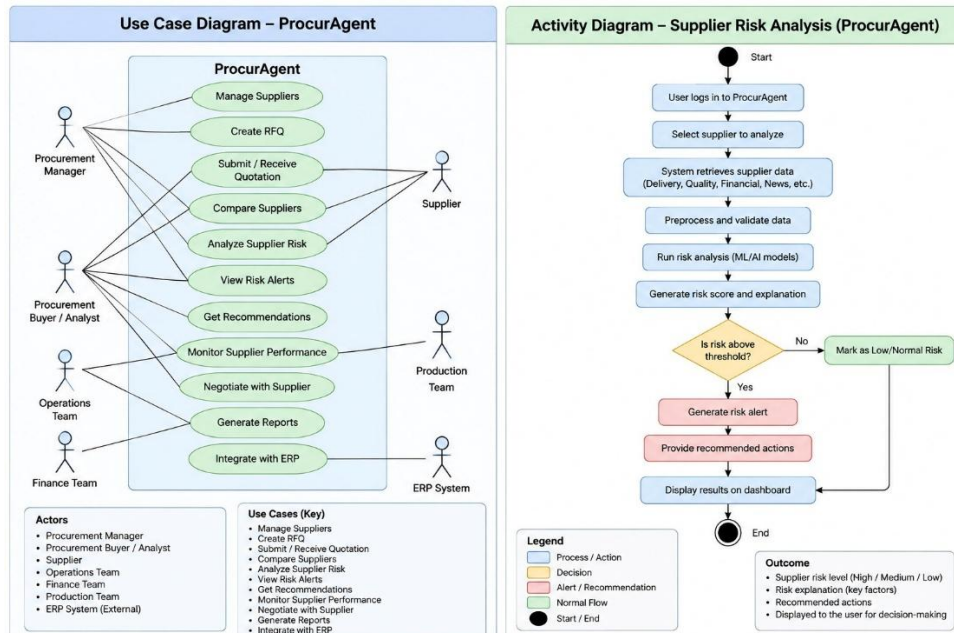
## 7. USE-CASE VIEW

### 7.1 ARCHITECTURALLY SIGNIFICANT USE CASES

| USE CASE                | MAIN ACTOR                     | ARCHITECTURAL PURPOSE              |
|-------------------------|--------------------------------|------------------------------------|
| View Dashboard          | Procurement Manager            | Aggregate procurement intelligence |
| Create RFQ              | Procurement Manager            | Start supplier procurement         |
| Process Quotation       | Buyer / Analyst                | Convert quote into structured data |
| Compare Suppliers       | Procurement Manager            | Multi-factor supplier evaluation   |
| Identify Risk           | Procurement Manager            | Predict and explain emerging risks |
| Generate Recommendation | Procurement Manager            | Evaluate procurement actions       |
| Negotiate               | Procurement Manager / Supplier | AI-assisted negotiation            |
| Approve Decision        | Manager / Finance              | Human governance                   |
| Monitor Supplier        | Procurement / Operations       | Continuous monitoring              |

## 7.2 Access Principle

Different roles see only the information and actions permitted to them.



## 8. LOGICAL / COMPONENT VIEW

## **8.1 MAJOR COMPONENTS**

### **Frontend**

- Dashboard
- Supplier Management
- RFQ Management
- Quotation Management
- Supplier Comparison
- Risk & Insights
- Negotiation
- Approvals
- Reports

### **FastAPI Backend**

- Authentication
- RBAC
- Supplier services
- RFQ services
- Quotation services
- Procurement services
- Risk services
- Recommendation services
- Negotiation services
- Database access

### **SNS Workbench**

- Master Procurement Agent
- Planning & Inventory
- Supplier Intelligence
- Risk & Monitoring
- Decision & Prescriptive
- Negotiation

## Python Services

- Data preprocessing
- Feature engineering
- ML inference
- Risk scoring
- Forecasting
- Price analysis
- Optimization

## PostgreSQL

Persistent structured procurement data.

# 9. COMPONENT RESPONSIBILITIES & INTERFACES

## COMPONENT PRIMARY RESPONSIBILITY

|                |                            |
|----------------|----------------------------|
| React/Vite     | User interface             |
| FastAPI        | APIs and backend services  |
| SNS Workbench  | AI-agent orchestration     |
| Python         | ML and analytics           |
| PostgreSQL     | Persistent structured data |
| External APIs  | External information       |
| Authentication | Identity and access        |
| Monitoring     | Events and alerts          |

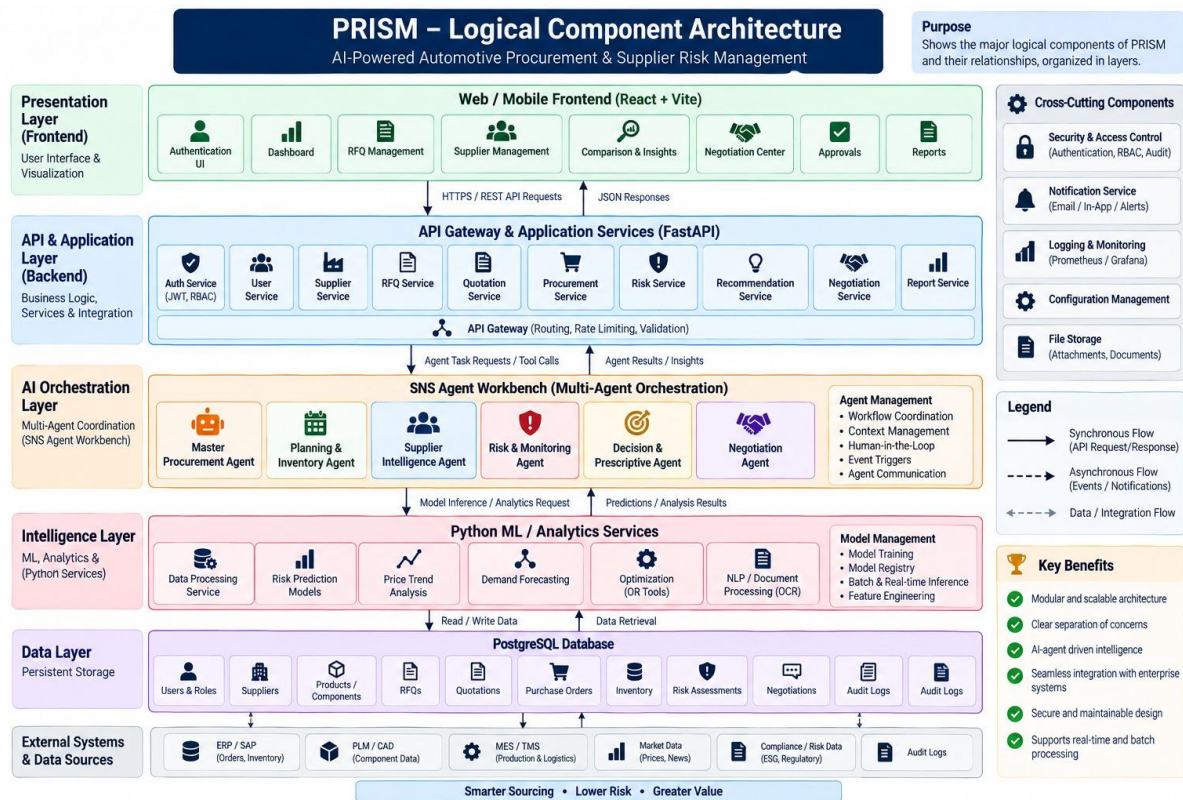
## Interface Principle

Frontend communicates with backend through APIs.

Backend manages:

**Authentication → Business Logic → Database → Agent Communication**

AI workflows do not directly expose database operations to the frontend.



The current SAD explicitly separates these responsibilities.

## 10. PROCESS / AGENT VIEW

### 10.1 MASTER PROCUREMENT AGENT

The Master Procurement Agent acts as the central orchestrator.

It:

- Receives procurement requests
- Understands context
- Allocates tasks
- Coordinates information flow
- Aggregates workflow outputs
- Handles new events
- Triggers re-analysis
- Provides consolidated insights

The Master Agent **coordinates** specialist workflows; it does not replace them.

### Specialist Workflows

1. Planning & Inventory
2. Supplier Intelligence
3. Risk & Monitoring
4. Decision & Prescriptive
5. Negotiation

## **11. PLANNING & SUPPLIER INTELLIGENCE WORKFLOWS**

### **11.1 PLANNING & INVENTORY WORKFLOW**

#### **Purpose**

**What do we need to procure?**

#### **Inputs**

- Production requirements
- BOM
- Current inventory
- Incoming supply
- Required dates

#### **Output**

**Net Requirement → Procurement Requirement**

Example:

Production Requirement + BOM + Current Inventory + Incoming Supply → Procurement Requirement

### **11.2 SUPPLIER INTELLIGENCE WORKFLOW**

#### **Purpose**

**Who can reliably provide what we need?**

#### **Evaluation Factors**

- Price
- Capacity
- Delivery performance
- Quality

- Lead time
- Historical performance
- Commercial conditions

### **Output**

Supplier comparison and suitability information.

## **12. RISK & MONITORING WORKFLOW**

### **Purpose**

**What can go wrong, and is anything starting to go wrong?**

### **Risk Categories**

- Delivery
- Material shortage
- Capacity
- Quality
- Supplier performance
- Commercial / Price

### **Monitoring Signals**

- Delivery postponements
- Lead-time variation
- Quality rejection
- Quantity shortages
- Capacity constraints
- Supplier status changes
- Inventory conditions
- Shipment/delivery milestones
- Supplier updates

### **Risk Principle**

A signal is not automatically proof of failure.

PRISM combines multiple signals and context to assess emerging risk.

## **Output**

**Risk Type + Score + Severity + Evidence + Confidence + Impact**

## **13. DECISION & PRESCRIPTIVE WORKFLOW**

### **Purpose**

#### **What should we do?**

The workflow combines information from:

- Planning
- Supplier Intelligence
- Risk & Monitoring
- Historical procurement data
- Business constraints

#### **Candidate Actions**

- Continue with supplier
- Negotiate delivery
- Split order
- Select alternative supplier
- Increase safety stock

#### **Evaluation Factors**

##### **Factor**

Cost

Risk

Delivery

Quality

Inventory

Production Impact

The workflow generates an explainable recommendation for human review.

# **14. NEGOTIATION WORKFLOW & AGENT ORCHESTRATION**

## **14.1 NEGOTIATION WORKFLOW**

### **Purpose**

**How can we negotiate better?**

### **Inputs**

- Current quotation
- Historical prices
- Supplier conditions
- Risk context
- Commercial terms

### **Outputs**

- Target price
- Counteroffer
- Negotiation strategy
- Message draft
- Supplier response analysis

## **14.2 COMPLETE AGENT FLOW**

### **Master Agent**

↓

**Planning + Supplier Intelligence + Risk & Monitoring**

↓

**Decision & Prescriptive**

↓

**Negotiation**

↓

**Human Approval**

↓

**Purchase Execution**

↓

## Continuous Monitoring

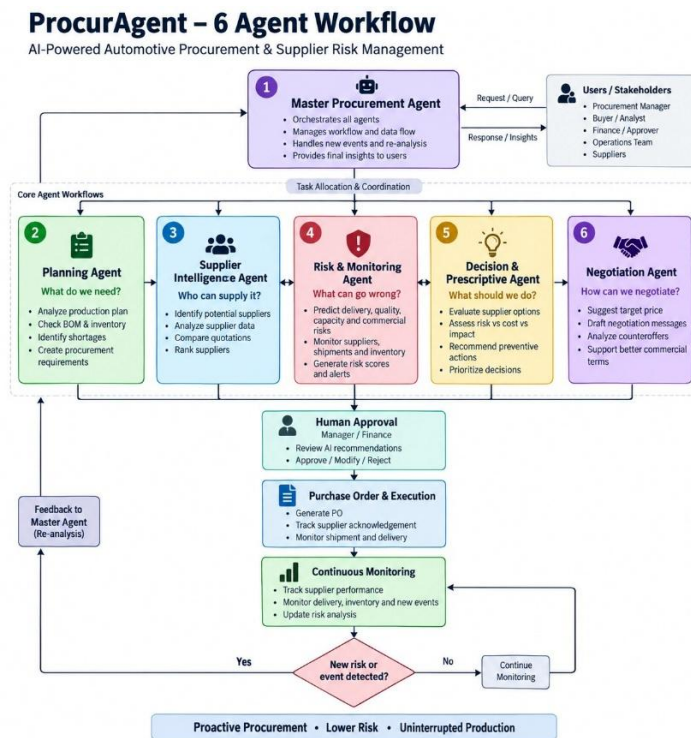


### New Event?

→ Yes → **Re-analysis** → **Master Agent**

→ No → **Continue**

This orchestration flow is already defined in the current SAD.



## 15. DEPLOYMENT VIEW

### 15.1 DEPLOYMENT ARCHITECTURE

PRISM is designed as a distributed application separating:

- Frontend
- Backend
- AI orchestration
- ML services
- Database

## **15.2 FRONTEND**

### **React/Vite**

- Web application
- HTTPS API communication
- No direct database access
- No direct internal AI-service access

## **15.3 BACKEND**

### **FastAPI**

Handles:

- Authentication
- Authorization
- APIs
- Business services
- Database access
- Agent communication
- External integrations

## **15.4 AI & DATA SERVICES**

### **SNS Workbench**

- Agent execution
- Routing
- Communication
- Event triggers
- Reasoning
- Human-in-the-loop

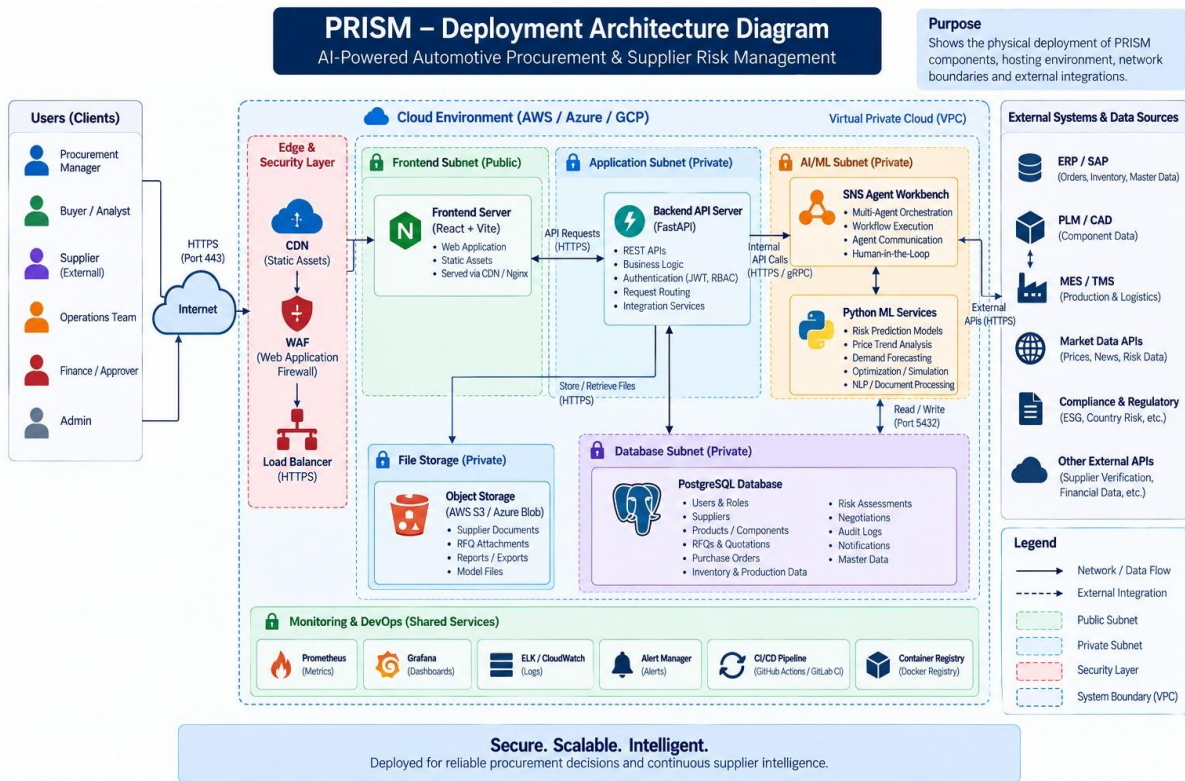
### **Python**

- Risk prediction
- Forecasting
- Price analysis
- Risk scoring

- Optimization

## PostgreSQL

- Persistent structured procurement data



## 16. DATA ARCHITECTURE

### 16.1 MAJOR DATA ENTITIES

- User
- Supplier
- Product / Component
- RFQ
- Quotation
- Purchase Order
- Inventory
- Historical Purchase
- Supplier Performance

- Risk Assessment
- Negotiation
- Recommendation
- Approval
- Shipment / Delivery

## 16.2 DATA RELATIONSHIPS

Supplier → Quotations → RFQ → Component

Supplier → Performance → Risk Assessment

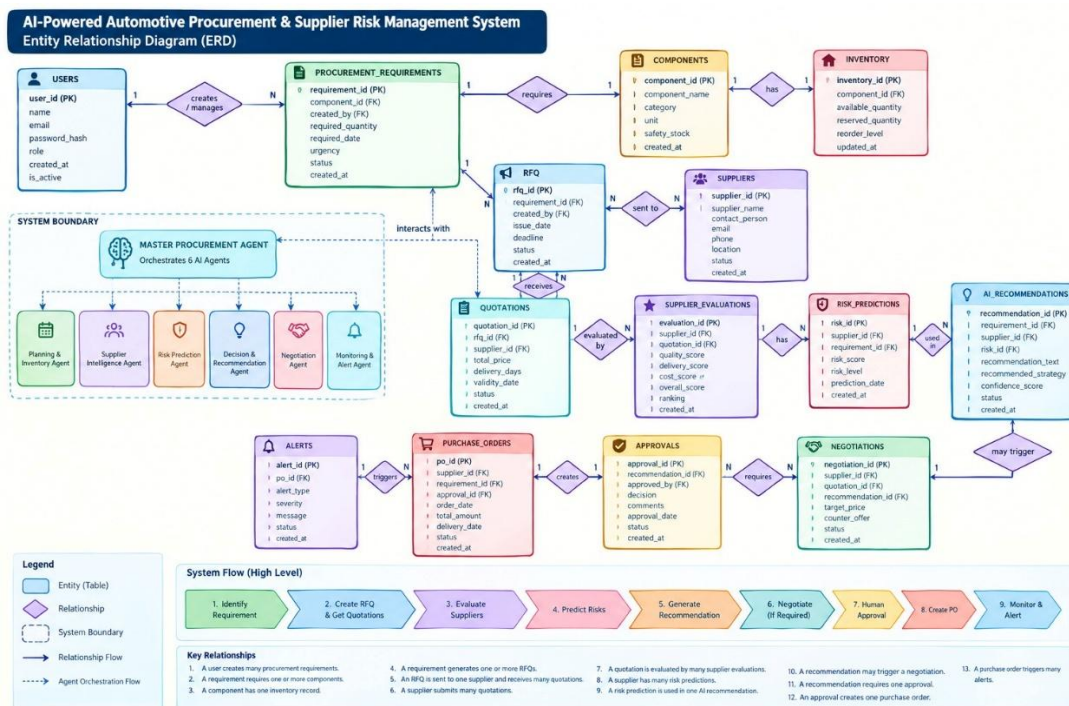
Component → Inventory → Procurement Requirement

Quotation → Historical Price → Negotiation

Risk + Procurement Context → Recommendation → Approval

## 16.3 DATA PRINCIPLE

PostgreSQL acts as the primary structured system of record.



## 17. SECURITY, RBAC & GOVERNANCE

## 17.1 SECURITY

PRISM uses:

- Authentication
- Role-Based Access Control
- Protected APIs
- Controlled data access
- Auditable decisions

## 17.2 ROLE-BASED ACCESS

| Role                | Main Access                                                          |
|---------------------|----------------------------------------------------------------------|
| Procurement Manager | Procurement, supplier, risk, recommendations, negotiation, approvals |
| Procurement Analyst | RFQ, quotation and procurement analysis                              |
| Supplier            | Own RFQs, quotations, negotiation and permitted updates              |
| Operations          | Supplier profile and qualification information                       |
| Finance             | Commercial and financial approval                                    |

## 17.3 GOVERNANCE RULES

- AI cannot automatically finalize important commercial decisions.
- Humans approve important recommendations.
- Missing information must not be fabricated.
- New supplier history must not be fabricated.
- Important decisions should be auditable.
- Supplier information must respect role permissions.

# 18. QUALITY ATTRIBUTES & TECHNOLOGY STACK

## 18.1 QUALITY ATTRIBUTES

**ATTRIBUTE    ARCHITECTURAL REQUIREMENT**

|                 |                                                  |
|-----------------|--------------------------------------------------|
| Performance     | Timely processing of procurement requests        |
| Security        | RBAC and protected information                   |
| Scalability     | Support additional suppliers, data and workflows |
| Reliability     | Consistent system and data behavior              |
| Explainability  | Evidence-based AI output                         |
| Maintainability | Modular services and workflows                   |
| Auditability    | Traceable important decisions                    |
| Usability       | Clear procurement insights                       |

**18.2 TECHNOLOGY STACK**

**LAYER            TECHNOLOGY**

|                  |                           |
|------------------|---------------------------|
| Frontend         | React + Vite              |
| Backend          | FastAPI                   |
| AI Orchestration | SNS Agent Workbench       |
| ML / Analytics   | Python                    |
| Database         | PostgreSQL                |
| Communication    | REST APIs                 |
| Authentication   | RBAC-based access control |

These quality goals and technology assignments are consistent with the current SAD.

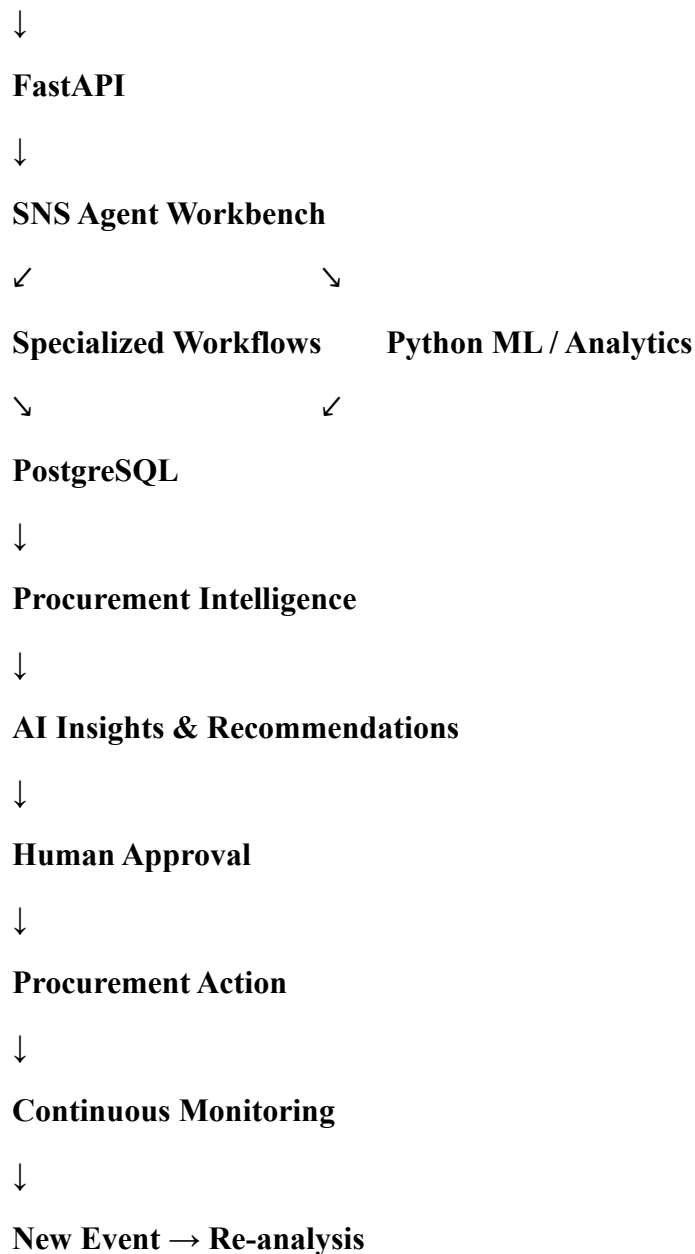
**19. FINAL ARCHITECTURE SUMMARY & REFERENCES**

**19.1 FINAL PRISM ARCHITECTURE**

**USERS**

↓

**React + Vite**



This is the condensed architectural flow already represented in the current SAD appendix.

## 19.2 REFERENCES

1. PRISM Product Requirements Document
2. PRISM System Design Documentation
3. PRISM Multi-Agent Workflow Design
4. PRISM Database / ER Design
5. Project Technical Requirements and Implementation Guidelines

## 19.3 APPENDIX

Only include diagrams that are **not already shown in the main sections**, such as:

- Activity Diagram
- Sequence Diagram
- Class Diagram
- AS-IS / TO-BE Process Diagram
- User Journey Map

The current SAD already identifies these as supporting diagrams.